Collections Copilot

Adda Wale | Implemented local build | Snapshot 28 September 2026

Source release: CRM, cards, lending, core banking, collections and bureau data. Immutable CSV / Parquet, English and French notes/transcripts, current policy documents.

1 Data Product Factory

DuckDB raw → curated → C360
Typed checks retain raw values and string IDs.
Cross-source identity rules; uncertain links held for review, not automatically merged.
Outputs: C360, lineage, counted quality report and YAML contract.

2 Insight / NLP

FastAPI + reviewed metric compiler
Local or optional Gemini metric/group/date plans → bound SQL.
Model-written SQL is rejected.
BM25 policy retrieval; exact, timestamped call statements. Refusals and clarification.
Outputs: cited answers and benchmark CSV.

3 Feature Store

One versioned offline/live definition
Structured features and bilingual financial cues.
Polars / Pandera checks. Parity measured on 100 customers: zero mismatches.
Outputs: stored features and source/version registry; classifier inputs unchanged.

4 Model / Decisioning

AI: scikit-learn logistic review classifier
Rules: holds → recorded hardship → possible cues.
Facts, reasons and draft; optional Gemini evidence selection cannot change decisions.
Human: supervisor approve / reject / edit.
Outputs: pending brief and SQLite review/audit.

Application: Next.js / Tailwind / Radix → FastAPI. Local DuckDB analytics; SQLite sessions/reviews. Gemini is optional and stateless; no key means local operation.

Governance across layers: server-issued roles; assigned-case access; source lineage and audit; protected fields excluded from serving/features. No automatic contact. Match scores are uncalibrated.